Sandeep Yadav

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Education

NIT Jalandhar **Aug 2024 – May 2028**

B.Tech, Computer Science & Engineering — CGPA: 7.89 / 10 (Top 20% of class) *Jalandhar, India*

- Relevant coursework: **Data Structures & Algorithms, Object-Oriented Programming, DBMS, Computer Networks, Operating Systems**.

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript, SQL

Web & Full-Stack: HTML, CSS, JavaScript, Full-Stack Web Development, Next.js, Node.js

Backend & APIs: RESTful APIs, Flask, HTTP, JSON, JWT Authentication, CRUD Operations, SQLAlchemy, Redis

Cloud & DevOps: Microsoft Azure, Azure App Service, Cloudflare Workers, Git, GitHub, Tauri

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems, Machine Learning

Experience

Digital Fire Control System (DFCS) — Freelance Project **Dec 2025 – Aug 2026**

Sole Developer *Demo*

- **Built** the core offline ballistic computation engine handling **4,000+ input parameters** as a cross-platform desktop application using **Tauri, JavaScript, and Python** (~**15,000 lines** across **10+ modules**).
- **Optimized** data processing pipeline to load and compute **100,000+ data rows in under 20ms**.
- **Translated** all ballistic calculation requirements gathered directly from **Indian Army stakeholders** into technical formulas and designed the full UI/UX, including interactive data visualizations.
- **Designed and implemented** a serverless auto-update pipeline using **Cloudflare Workers**, with separate staging and production environments, to push signed application updates to field devices safely.
- Delivered the system with **100% accuracy** against all Army-provided test cases.

Projects

Stonks | *AI-Powered Investment Intelligence Platform* **GitHub**

- **Developed** a full-stack AI-agentic dashboard using **Next.js, Node.js, LangChain, RAG, and ChromaDB**, synthesizing real-time insights from live news feeds and SEC filings.
- **Pioneered** vector memory and semantic retrieval through **ChromaDB** to enhance multi-turn financial Q&A, leading to a measurable **40% improvement** in answer relevance compared to baseline LLM prompts.
- **Established** secure SSR routes and API endpoints with **JWT authentication**, refresh-token rotation, and role-aware access control.

Bio-Shield | *AI-Based Behavioral Biometrics for UPI Security* **GitHub**

- **Developed** a **Flask**-based behavioral-biometric security system for **UPI payments**, capturing **keystroke dynamics** to continuously authenticate users during a transaction session.
- **Implemented** a real-time risk engine using **weighted Z-score analysis** to classify transactions as **ALLOW, OTP_REQUIRED, or BLOCK**, with **OTP** and **OpenCV/DeepFace** face-verification fallback authentication.
- **Engineered** the backend with **SQLAlchemy, PostgreSQL/SQLite, Redis, JWT, and bcrypt** for secure session management, password hashing, and low-latency risk scoring.

Research Document Dashboard | *JavaScript, HTML, CSS, Azure App Service* **GitHub**

- **Independently designed and deployed** a full-stack web dashboard on **Microsoft Azure**, serving **50+ academic documents**, with dynamic client-side filtering that cut retrieval time by ~**70%**.
- **Took** the project from local development to a live Azure deployment in a single day, with a responsive UI for desktop and mobile.

AgriMarket Profit Optimiser | *Python, Pandas, NumPy, Optimisation* **GitHub**

- **Built** a Python decision-support tool that **analyzes multi-market crop pricing data** and **implements an optimization algorithm** to recommend the most profitable markets for selling crops, replacing manual comparison with a data-driven workflow.

Achievements & Certifications

- **Microsoft Azure** — Hands-on cloud deployment via Azure App Service (December 2025).
- **LeetCode** — **400+ problems solved** across Arrays, Linked Lists, Trees, Dynamic Programming, and Graphs (2024–Present).
- **Codeforces** — Rated competitive programmer, **5+ rated contests** completed (Handle: sandeepy.cs.24).
- **Mobile Security Bootcamp** (IIT Jammu × NIT Jalandhar) — **2nd Place**, 2025.
- **Devcation, IIT Delhi** — Semifinalist, 2025.
- **Nerds Room Hackathon** — Semifinalist, 2025.